2018 Ph H2 Q2

Section: Our Dynamic Universe

Topic: Forces, power, tension in cables

(a)(i) A drone of mass 5.50 kg carries a 1.25 kg package while hovering. Find the upward force produced.
(ii) The package is lowered by a motor (12 V across 6 Ω). Calculate the power dissipated.
(iii) If the cable breaks but the upward thrust stays the same, describe the drone's vertical motion.

(b) Two drones share a package of mass 3.4 kg using two cables each at 35°. Find the tension in each cable.

Worked solution

(a)(i)
Total mass = 5.50 + 1.25 = 6.75 kg.
Weight = mg = 6.75 × 9.8 = 66.2 N.

Answer: 66 N

(a)(ii)
$P = V^2/R = 12^2/6 = 24.0$ W.

Answer: 24 W

(a)(iii)
With the package gone, the upward thrust is greater than the drone's own weight. So there is a net upward force. The drone will accelerate vertically upwards

immediately.

Answer: Drone accelerates upward

(b)
Weight = 3.4 × 9.8 = 33.3 N.
Two tensions support this vertically: $2T \sin 35° = W$.
$T = W / (2 \sin 35°) = 29.0$ N.

Answer: 29 N

Final answers

(a)(i) 66 N

(a)(ii) 24 W

(a)(iii) Drone accelerates upward

(b) Tension ≈ 29 N

Revision tips

- To hover: thrust = total weight (drone + package).
- Electrical power in a resistor: $P = V^2 / R$.
- If thrust > weight → upward acceleration.
- When two cables share weight at angle θ: $2T \sin\theta = W$.
- Check vector balance carefully in 2D force problems.